AI Powered Quantitative Trading Fundamentals

Course code C728

2 days | 15 instructional hours

LESSON PLAN

Tertiary Infotech Academy Pte Ltd

UEN 201200696W

Version 2.1 | 25 October 2026

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Version History

Version No	Date Effected	Change Update	Approved by
2.1	25 October 2026	Lab 20 capstone now runs the trading simulator's eight-agent workflow (lead, finance research, news sentiment, trading strategy, signal generation, stock trader, performance analysis, strategy review) with its feedback loop; Idea > Strategy step; new Lab 20 screenshot.	Dr Alfred Ang
2.0	24 October 2026	Release of AI Powered Quantitative Trading Fundamentals (C728): two days, 9:30am - 5:30pm, 15 instructional hours; four topics and twenty hands-on labs in the low-code Trading Simulator.	Dr Alfred Ang

This document is at Version 2.1, effective 25 October 2026.

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1. Course Information

Item	Details
Course Title	AI Powered Quantitative Trading Fundamentals
Course Code	C728
Course Duration	15 hours over 2 training days
Daily Schedule	9:30am - 5:30pm (7.5 instructional hours per day, excluding a 30-minute lunch break)
Mode of Delivery	Instructor-led classroom training with hands-on practical laboratory exercises on the Alpaca paper-trading platform, TradingView and the low-code Trading Simulator
Software Platforms	Alpaca paper-trading account (Trading API and Market Data API, alpaca-py SDK); TradingView (charts, Pine Script v6, Strategy Tester, alerts); the course's low-code Trading Simulator (https://github.com/alfredang/algorithmic-trader-bot)
Training Venue	12 Woodlands Square, #07-85/86/87, Woods Square Tower 1, Singapore 737715
Class Size	Optimum class size of 20 participants (maximum 20)
Trainer to Learner Ratio	1 : 20

2. Target Audience and Entry Requirements

Target Audience

- Traders, dealers, investment and research analysts in banks, brokerages and asset managers who want to test and automate trading ideas with data and machine learning.
- Quantitative, risk and data analysts in financial institutions who support trading, treasury or investment functions.
- Finance executives and technology professionals moving into algorithmic trading, trading technology or AI roles in financial services.
- Individual investors working in financial services who want a disciplined, evidence-based approach to systematic trading.

Entry Requirements

Knowledge and Skills

- Able to read, write, speak and listen in English at a proficiency level equivalent to the Employability Skills Workplace Literacy Level 5.
- Able to manipulate numbers at a proficiency level equivalent to the Employability Skills Workplace Numeracy Level 5.
- Basic computer literacy; no programming is required - every lab runs in a low-code trading simulator.
- Basic understanding of financial markets (stocks, ETFs, buying and selling).

Experience

- Preferably 1 year of working experience in financial services or a related role involving data, investment or trading.

Equipment

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- A laptop (macOS, Windows or Linux) with a modern web browser and internet access, able to run the free course Trading Simulator (installation is guided in Lab 1).
- A free Alpaca paper-trading account and a free TradingView account, created in class; no real money is used at any point.

3. Learning Outcomes

On completion of this course, participants will be able to:

LO	Learning Outcome
LO1	Develop hypotheses for implementing data analytics and computational modelling activities in algorithmic trading, and identify possible big data applications across the forms of data within financial services.
LO2	Identify appropriate statistical algorithms and data models to test trading hypotheses, and use appropriate analytics platforms, analytical tools and statistical methods given specific analytics and reporting requirements.
LO3	Conduct statistical modelling of market data with machine learning models to derive patterns and solutions, and perform A/B testing of trading strategies based on different hypotheses.
LO4	Perform coding and configuration of an automated trading program based on selected models or algorithms, conduct tests on its actions and outcomes, diagnose unintended outcomes, and draw insights from the results to support decisions and report accordingly.

4. Course Structure and Duration by Topic

Topic	Title	Day	Labs	Duration	Slides
Topic 1	Foundations of Quant and Algorithmic Trading: From Thorp to AI Agents	Day 1	Labs 1-4	2 h 55 min	26 - 71
Topic 2	Technical Indicators, Trading Strategies and Metrics with TradingView and Alpaca	Day 1	Labs 5-9	3 h 50 min	72 - 142
Topic 3	Machine Learning Models, Statistical Modelling and A/B Testing for Trading Signals	Day 2	Labs 10-14	3 h 10 min	143 - 182
Topic 4	Building an End-to-End Algorithmic Trading Pipeline: Backtest, Forward Test and Performance Analysis	Day 2	Labs 15-20	4 hours	183 - 213
Admin	Course overview, ground rules, daily recaps	1, 2	-	1 h 5 min	1 - 25
TOTAL				15 hours	

Topic hours include the trainer-led content and the hands-on labs for that topic. 15 instructional hours in total over 2 days, excluding the lunch breaks.

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Hands-on labs

Lab	Title	Topic	Duration	Slide
1	Pipeline Skeleton, Configuration and the Alpaca Paper Account	1	20 min	58
2	Data-Validation Stage: Cross-Check Pipeline Bars against TradingView	1	20 min	61
3	Ingestion Stage: Multi-Timeframe Bars, Headlines and Data Quality	1	20 min	64
4	Hypothesis Stage: from a Written Card to an Automated Statistical Test	1	15 min	68
5	Indicator Stage: Pine Script v6 Indicators and their Python Twins	2	25 min	119
6	Strategy Prototype: TradingView Strategy Tester, Paper Trading from the Chart and Pipeline Reconciliation	2	25 min	123
7	Signal Stage: a Library of Trend, Mean-Reversion, Momentum and Breakout Signals	2	25 min	128
8	Execution Stage: an Order Manager that Trades the Paper Account in Any Session	2	25 min	132
9	Evaluation Stage: Performance and Risk Metrics with a Benchmark Table	2	25 min	137
10	Strategy-Search Stage: Grid Search, Out-of-Sample Ranking and Decay	3	25 min	165
11	ML-Signal Stage: a Walk-Forward Classifier from Probability to Position	3	25 min	169
12	Regime-Filter Stage: PCA and K-Means as a Gate on the Signals	3	20 min	172
13	A/B-Test Stage: a Statistical Decision before Capital Moves	3	20 min	175
14	Pairs Stage: a Hedged, Market-Neutral Statistical-Arbitrage Sleeve	3	25 min	179
15	Backtest-Engine Stage: Cost Models, Execution Timing and Break-Even	4	30 min	195
16	Automation Stage: a Scheduled, Risk-Checked Forward Test and Live Paper Trading	4	30 min	198
17	Event-Driven Automation: TradingView Webhook to Signal, Risk and Execution	4	25 min	201
18	Diagnostics Stage: Automated Bias Checks before Go-Live	4	25 min	204
19	Monitoring and Reporting Stage: Automated Tearsheet and Backtest-vs-Forward Report	4	30 min	207
20	Capstone: One Command Runs the Pipeline End to End, with a Multi AI Agents Workflow	4	25 min	210

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5. Daily Training Schedule

Each training day runs from 9:30am to 5:30pm: 7.5 instructional hours, excluding a 30-minute lunch break.

Day 1

Time	Topic / Activity	Learning Outcome	Delivery Method & Resources
09:30 - 10:00	Course Overview, Trainer Introduction and Ground Rules Course objectives, ground rules and an icebreaker on how participants use data in trading or investment today.	-	Trainer presentation; Slides; Learner Guide
10:00 - 11:10	T1.1-1.4 Quant Success Stories, the Evolution of Quant, the ML-to-LLM Shift, and Agentic AI versus AI Agents Quant success stories (Thorp, Renaissance); the evolution of quant to high-frequency trading and the six paradigm shifts; machine learning to LLMs and the impact on quant; agentic AI versus AI agents and multi-agent trading systems.	LO1	Trainer presentation, discussion and worked examples; Slides; Learner Guide
11:10 - 11:40	T1.5-1.6 Forms of Financial Data, Big Data Applications and Trading Hypotheses Market, reference, fundamental and alternative data; the four Vs; big data applications; writing a testable hypothesis.	LO1	Trainer presentation, discussion and worked examples; Slides; Learner Guide
11:40 - 12:00	Lab 1: Pipeline Skeleton, Configuration and the Alpaca Paper Account Get the simulator running, decide between the offline demo and an Alpaca paper account, and check every integration's status.	LO1	Hands-on lab in the Trading Simulator; Learner Guide; lab manual Lab01_Instructions.pdf
12:00 - 12:20	Lab 2: Data-Validation Stage: Cross-Check Pipeline Bars against TradingView Prove the app's prices are right: compare its bars with TradingView's Data Window and explain adjusted vs unadjusted prices.	LO1	Hands-on lab in the Trading Simulator; Learner Guide; lab manual Lab02_Instructions.pdf
12:20 - 12:40	Lab 3: Ingestion Stage: Multi-Timeframe Bars, Headlines and Data Quality Ingest bars from Alpaca / Twelve Data / yfinance (or the demo files), run the data-quality checks and score news sentiment.	LO1	Hands-on lab in the Trading Simulator; Learner Guide; lab manual Lab03_Instructions.pdf
12:40 - 13:10	Lunch Break		
13:10 - 13:25	Lab 4: Hypothesis Stage: from a Written Card to an Automated Statistical Test Write H0 and H1 before looking at results and test them with a Welch t-test and a bootstrap CI.	LO1	Hands-on lab in the Trading Simulator; Learner Guide; lab manual Lab04_Instructions.pdf
13:25 - 14:05	T2.1-2.2 Technical Indicators and Standard Algorithmic Trading Strategies SMA, EMA, RSI, MACD, Bollinger Bands, ATR, VWAP, OBV; trend following, mean reversion, momentum, breakout, pairs trading and execution algorithms.	LO2	Trainer presentation, discussion and worked examples; Slides; Learner Guide
14:05 - 14:30	Lab 5: Indicator Stage: Pine Script v6 Indicators and their Python Twins Plot SMA/EMA, Bollinger Bands, RSI and MACD and export a Pine Script v6 strategy for TradingView.	LO2	Hands-on lab in the Trading Simulator; Learner Guide; lab manual Lab05_Instructions.pdf

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Time	Topic / Activity	Learning Outcome	Delivery Method & Resources
14:30 - 14:55	Lab 6: Strategy Prototype: TradingView Strategy Tester, Paper Trading from the Chart and Pipeline Reconciliation Run the same MA-crossover in TradingView's Strategy Tester and in the app, reconcile the results, and try TradingView Paper Trading.	LO2	Hands-on lab in the Trading Simulator; Learner Guide; lab manual Lab06_Instructions.pdf
14:55 - 15:20	Lab 7: Signal Stage: a Library of Trend, Mean-Reversion, Momentum and Breakout Signals Pick and customise one strategy from each family and compare them after costs.	LO2	Hands-on lab in the Trading Simulator; Learner Guide; lab manual Lab07_Instructions.pdf
15:20 - 16:00	T2.3-2.4 Analytics Platforms, Coding Languages and Order Placement TradingView, Alpaca Trading and Market Data APIs, Python libraries; Pine Script and Python; order types, time in force, bracket orders and the order lifecycle.	LO2	Trainer presentation, discussion and worked examples; Slides; Learner Guide
16:00 - 16:25	Lab 8: Execution Stage: an Order Manager that Trades the Paper Account in Any Session Place market, limit, stop and bracket paper orders, including an extended-hours limit order, and follow each order's lifecycle.	LO2	Hands-on lab in the Trading Simulator; Learner Guide; lab manual Lab08_Instructions.pdf
16:25 - 16:50	T2.5 Performance and Risk Metric Definitions CAGR, volatility, Sharpe, Sortino, Calmar, maximum drawdown, win rate, profit factor, expectancy, exposure, turnover, alpha and beta.	LO2	Trainer presentation, discussion and worked examples; Slides; Learner Guide
16:50 - 17:15	Lab 9: Evaluation Stage: Performance and Risk Metrics with a Benchmark Table Read the full metrics table and compare a strategy with buy-and-hold.	LO2	Hands-on lab in the Trading Simulator; Learner Guide; lab manual Lab09_Instructions.pdf
17:15 - 17:30	Day 1 Recap and Q&A Consolidation of Topics 1 and 2; preview of Day 2.	-	Trainer presentation; Slides; Learner Guide

Total instructional time, Day 1: 7.5 hours (450 minutes), excluding the 30-minute lunch break.

Day 2

Time	Topic / Activity	Learning Outcome	Delivery Method & Resources
09:30 - 09:50	Day 1 Recap and Day 2 Overview Review of Topics 1 and 2; orientation to Topics 3 and 4.	-	Trainer presentation; Slides; Learner Guide
09:50 - 10:20	T3.1-3.2 Statistical Analyses, Algorithms and Supervised Learning Models Descriptive to prescriptive analysis; supervised, unsupervised and reinforcement learning; strategy search; regression, trees, random forest, XGBoost, neural networks.	LO3	Trainer presentation, discussion and worked examples; Slides; Learner Guide
10:20 - 10:45	Lab 10: Strategy-Search Stage: Grid Search, Out-of-Sample Ranking and Decay Grid-search SMA crossover parameters in-sample and measure how the ranking decays out of sample.	LO3	Hands-on lab in the Trading Simulator; Learner Guide; lab manual Lab10_Instructions.pdf
10:45 - 11:10	Lab 11: ML-Signal Stage: a Walk-Forward Classifier from Probability to Position Train logistic regression, random forest and XGBoost walk-forward and trade P(up) > threshold.	LO3	Hands-on lab in the Trading Simulator; Learner Guide; lab manual Lab11_Instructions.pdf

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Time	Topic / Activity	Learning Outcome	Delivery Method & Resources
11:10 - 11:35	T3.3-3.4 Unsupervised Learning and Statistical Modelling Techniques K-means, hierarchical clustering and PCA for regimes; returns and stationarity; time-ordered validation; feature engineering; regularisation.	LO3	Trainer presentation, discussion and worked examples; Slides; Learner Guide
11:35 - 11:55	Lab 12: Regime-Filter Stage: PCA and K-Means as a Gate on the Signals Find market regimes with walk-forward K-means and gate a signal in the high-volatility regime.	LO3	Hands-on lab in the Trading Simulator; Learner Guide; lab manual Lab12_Instructions.pdf
11:55 - 12:15	T3.5-3.6 A/B Testing, and Pairs Trading as a Hedging Algorithm Hypotheses, Welch's t-test, bootstrap Sharpe difference, multiple testing; cointegration, hedge ratio, spread z-score and market neutrality.	LO3	Trainer presentation, discussion and worked examples; Slides; Learner Guide
12:15 - 12:35	Lab 13: A/B-Test Stage: a Statistical Decision before Capital Moves Decide statistically between two variants before moving capital.	LO3	Hands-on lab in the Trading Simulator; Learner Guide; lab manual Lab13_Instructions.pdf
12:35 - 13:00	Lab 14: Pairs Stage: a Hedged, Market-Neutral Statistical-Arbitrage Sleeve Screen pairs for cointegration and backtest a market-neutral KO/PEP spread.	LO3	Hands-on lab in the Trading Simulator; Learner Guide; lab manual Lab14_Instructions.pdf
13:00 - 13:30	Lunch Break		
13:30 - 14:10	T4.1-4.2 Pipeline Architecture and Backtesting with Trading Costs The eight-stage pipeline; configuration and pre-trade risk controls; next-bar execution; commission, spread, slippage and regulatory fees.	LO4	Trainer presentation, discussion and worked examples; Slides; Learner Guide
14:10 - 14:40	Lab 15: Backtest-Engine Stage: Cost Models, Execution Timing and Break-Even Run the same strategy under four cost models and find its break-even cost.	LO4	Hands-on lab in the Trading Simulator; Learner Guide; lab manual Lab15_Instructions.pdf
14:40 - 14:55	T4.3 Forward Testing with Automated Paper Trading The bot loop, market clock, risk checks, order status and logging.	LO4	Trainer presentation, discussion and worked examples; Slides; Learner Guide
14:55 - 15:25	Lab 16: Automation Stage: a Scheduled, Risk-Checked Forward Test and Live Paper Trading Create a bot, run it on a schedule, watch orders, positions and P&L live, and halt it with the kill switch.	LO4	Hands-on lab in the Trading Simulator; Learner Guide; lab manual Lab16_Instructions.pdf
15:25 - 15:50	Lab 17: Event-Driven Automation: TradingView Webhook to Signal, Risk and Execution Send TradingView alerts to the app's webhook and see them pass validation, signal confirmation and risk into paper orders.	LO4	Hands-on lab in the Trading Simulator; Learner Guide; lab manual Lab17_Instructions.pdf
15:50 - 16:00	T4.4 Potential Reasons for Unintended Outcomes Overfitting, look-ahead and survivorship bias, cost drag, regime change, data and software errors; the diagnosis method.	LO4	Trainer presentation, discussion and worked examples; Slides; Learner Guide
16:00 - 16:25	Lab 18: Diagnostics Stage: Automated Bias Checks before Go-Live Run the automated diagnostics on a deliberately biased strategy, fix it, and re-run.	LO4	Hands-on lab in the Trading Simulator; Learner Guide; lab manual Lab18_Instructions.pdf
16:25 - 16:35	T4.5 Performance Analysis, Insights and Reporting Tearsheets, backtest versus forward test, insights and the recommendation.	LO4	Trainer presentation, discussion and worked examples; Slides; Learner Guide

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Time	Topic / Activity	Learning Outcome	Delivery Method & Resources
16:35 - 17:05	Lab 19: Monitoring and Reporting Stage: Automated Tearsheet and Backtest-vs-Forward Report Produce a one-page tearsheet and explain the gap between backtest and forward test.	LO4	Hands-on lab in the Trading Simulator; Learner Guide; lab manual Lab19_Instructions.pdf
17:05 - 17:30	Lab 20: Capstone: One Command Runs the Pipeline End to End, with a Multi AI Agents Workflow Run the whole pipeline end to end, run the 8-agent workflow (research, sentiment, strategy, signal, paper trade, performance analysis, strategy review) and watch its feedback loop, switch on the always-on agent, and turn your own idea into a strategy, then backtest and forward-test it.	LO4	Hands-on lab in the Trading Simulator; Learner Guide; lab manual Lab20_Instructions.pdf

Total instructional time, Day 2: 7.5 hours (450 minutes), excluding the 30-minute lunch break.

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6. Training Resources

Trainer Resources

- Trainer slide deck (PowerPoint), 216 slides covering all four topics and twenty labs.
- Lesson Plan (this document).
- Lab folder (labs) with the twenty lab manuals; the Trading Simulator installed on the trainer laptop; a trainer Alpaca paper account for live demonstrations.
- Laptop with the Trading Simulator, a modern web browser, projector and whiteboard.

Learner Resources

- Learner Guide containing all topic content and step-by-step instructions for every lab.
- One laptop per participant with a modern web browser and internet access; a free Alpaca paper-trading account and a free TradingView account (created in Lab 1 and Lab 2).
- The lab folder with the twenty lab manuals (LabNN_Instructions.pdf).
- The course Trading Simulator (https://github.com/alfredang/algorithmic-trader-bot): a local web app (uv or Docker Compose) with a Course Labs page that walks through all twenty labs - no coding is needed.

7. Trainer Notes

- Paper trading only. Never demonstrate with a live-money account, and remind participants that nothing in the course is investment advice.
- If the venue network blocks the Alpaca API, or a participant cannot open an account, every lab runs offline on the bundled data and the built-in paper-broker simulator. Show both paths early (Lab 1) so nobody is blocked.
- This is a low-code course: run every lab in the Trading Simulator (Course Labs page, quick and visual). No participant needs to write code. Participants who want full customisation in Python can be pointed to the separate Full Code edition. Have the simulator installed and seeded on the trainer machine before Day 1 so you can project each lab's page.
- Webhook alerts on TradingView need a paid plan. In Lab 17 demonstrate with the trainer's account if available, and let participants use the simulator's 'Run self-test (7 simulated alerts)', which sends the identical JSON a TradingView alert sends.
- Lab 18 contains deliberately seeded faults. Do not reveal them; let participants find the look-ahead first, because it produces no error message - only a result that is too good.
- Keep the finance context foremost. When a participant reports a great backtest, ask the three questions: is it net of costs, is it out of sample, and what would the investment committee need to see before risking capital?
- In Topic 1, link every milestone of the AI and algorithmic-trading histories to a control used later in the course (circuit breakers to the kill switch, Knight Capital to deployment testing, MiFID II to algorithm records).

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